Smashifix: An Application for Badminton Posture Correction Using Deep Learning

Sian Rodrigues[1], Yati Rathod[2], Vedant Gadge[3], Smayan Kulkarni[4], Namita Pulgam[5], Pradnya Saval[6]

[1]Dept. of Computer Science and Engineering (Data Science), Dwarkadas J. Sanghvi College of Engg., Mumbai, India.

Contributing authors: Sianrods250@gmail.com; yatirathod01@gmail.com; vedant.gadgegsis@gmail.com; smayankulkarni05@gmail.com; Namita.Pulgam@djsce.ac.in; Pradnya.saval@djsce.ac.in;

Abstract

Badminton is a high-speed racquet sport demanding precise biomechanics and rapid reflexes, where even minor deviations in stroke technique can lead to stagnated performance and chronic injury. Traditional coaching, reliant on subjective observation, is inconsistent, expensive, and inaccessible to amateur players; existing automated solutions either classify actions without corrective feedback or require expensive multi-sensor hardware. This paper presents "Smashifix," an automated computer vision framework that addresses these gaps through a dual-pipeline architecture: a Pose-LSTM pipeline for real-time feedback (∼50ms latency) and an Attention-enhanced Temporal Convolutional Network (Attention-TCN) fusing skeletal data with visual context from a custom Residual-Shuffle Network (RSN) for forensic analysis (∼85% accuracy). A Kinematic Similarity Index (KSI) quantifies user performance against expert templates using Dynamic Time Warping (DTW), which a Natural Language Coach subsequently translates into personalized, skill-appropriate textual feedback. The framework classifies six distinct strokes and provides actionable biomechanical feedback using only a standard smartphone camera, enabling accessible, objective coaching for players at all levels.

Keywords: Markerless Pose Estimation, Attention-enhanced TCN, Residual-Shuffle Network, Kinematic Similarity Index, Biomechanical Feedback, Dynamic Time Warping, Natural Language Coach

1 Introduction

1.1 Background

The intersection of computer vision and artificial intelligence has fundamentally transformed the landscape of sports analytics, shifting the paradigm from subjective observation to data-driven decision-making. Modern systems possess the capability to track intricate movement patterns, providing athletes and coaches with objective, quantitative data that was previously inaccessible. In high-performance sports, even minor deviations in technique can significantly impact competitive outcomes. Badminton, in particular, is a sport defined by rapid reflexes, explosive power, and precise body mechanics. It requires a complex interplay of footwork, stroke execution, and tactical positioning, where the shuttlecock can travel at speeds exceeding 400 km/h. Consequently, the ability to analyze movement in granular detail—down to the angle of the wrist or the rotation of the hip—is essential for player development and injury prevention.

Historically, biomechanical analysis was confined to laboratory settings equipped with marker-based motion capture systems. While highly accurate, these systems are intrusive, expensive, and restricted to controlled environments, making them unsuitable for on-court coaching. The advent of markerless pose estimation algorithms has opened new avenues for "in-the-wild" analysis, allowing for the extraction of kinematic data from standard video feeds. This democratization of technology presents a unique opportunity to bring professional-grade analytics to amateur and semi-professional players.

1.2 Current Limitations and System Objectives

Despite advancements in sports technology, badminton coaching largely remains a manual and subjective process. Players often face challenges with improper posture and technique, which not only hampers skill development but also significantly increases the risk of chronic injuries such as lateral epicondylitis (tennis elbow) and patellar tendinopathy. Current automated solutions often suffer from critical limitations that hinder their widespread adoption:

- **Hardware Dependency:** Many high-fidelity systems rely on expensive wearable sensors (IMUs) or multi-camera setups. These are impractical for widespread use due to their cost and setup complexity.
- **Lack of Corrective Feedback:** Most existing computer vision applications focus solely on action recognition (e.g., classifying a smash vs. a drop) or basic statistical tracking. They fail to provide the "how" and "why" of a mistake, lacking the corrective feedback necessary for motor learning.
- **Latency vs. Accuracy Trade-off:** Current approaches often fail to balance the need for real-time processing with the depth required for forensic biomechanical analysis. Real-time models are often too lightweight to capture subtle nuances, while high-accuracy models are too computationally intensive for live feedback.

- **Robustness of Spatial Tracking:** Dynamic bounding box methods based on joints frequently suffer from temporal jitter, occlusion-induced failures, and inconsistent scaling, leading to noisy downstream feature extraction that compromises forensic analysis.

To bridge these gaps, the system “Smashifix” is proposed as an accessible, single-camera coaching assistant. The primary objectives of this research are:

1. To develop a hardware-agnostic framework that provides immediate, actionable feedback on player posture using only a standard smartphone camera.
2. To integrate lightweight deep learning models (Pose-LSTM) for real-time cues with robust hybrid models (Attention-TCN with RSN) for detailed forensic analysis.
3. To mathematically quantify deviations from expert templates using an enhanced Kinematic Similarity Index (KSI), thereby reducing the risk of injury through automated, objective correction.
4. To overcome the limitations of dynamic bounding box tracking by employing a robust, configuration-driven static spatial cropping methodology that guarantees consistent player scaling.

The organization of this paper is as follows: Section 2 presents the Literature Survey, reviewing state-of-the-art methods in pose estimation and sports analytics, followed by a comparative analysis. Section 3 outlines the proposed system, detailing the novel dual-pipeline architecture. Section 4 provides an in-depth explanation of the methodology, including data preprocessing via deterministic spatial cropping, mathematical formulations for the KSI, and feature engineering. Section 5 presents the experimental setup, dataset description, evaluation metrics, and a comprehensive discussion of the results. Finally, Section 6 concludes the study and outlines directions for future research.

2 Literature Survey

2.1 Survey of Existing Systems and Datasets

The development of automated sports analysis relies heavily on the availability of high-quality, domain-specific data and robust architectures.

Wang et al. [1] detailed a structured approach to creating a tennis player dataset, capturing 2,000 annotated frames of key actions like forehands and serves using COCO-Annotator. This study emphasized the critical importance of diverse pose representation for training robust models capable of generalizing across different athletes. In the specific domain of badminton, Zhang et al. [2] developed a human-computer interactive teaching system using Part Affinity Fields (PAF) trained on a massive dataset of 35,000 movements. While their system was effective for basic movement tracking and hit-point detection, it primarily functioned as a retrieval system rather than a granular biomechanical correction tool.

A comprehensive overview of these Human Pose Estimation (HPE) techniques is provided by Dibenedetto et al. [3]. Their survey categorizes methods into skeleton-based, surface-based, and volume-based models, highlighting the evolution from regression-based methods to modern detection frameworks like MediaPipe and OpenPose. They note that while 2D estimation has matured, persistent challenges remain in handling self-occlusion and crowded scenes, which are common in doubles badminton matches.

2.2 Pose Estimation Architectures

Accurate localization of joints is the cornerstone of biomechanical analysis. Cao et al. [4] established the viability of real-time 2D pose estimation with OpenPose, using Part Affinity Fields to associate keypoints in multi-person scenarios. This bottom-up approach was revolutionary but computationally expensive. Building on this, Lugaresi et al. [5] introduced MediaPipe, a modular framework optimized for on-device machine learning. Within this ecosystem, Bazarevsky et al. [6] proposed BlazePose, a lightweight detector-tracker architecture that predicts 33 body landmarks, making it ideal for real-time mobile applications where latency is a critical constraint.

Historically, Toshev and Szegedy [7] pioneered the use of Deep Neural Networks (DNNs) for pose estimation with DeepPose, shifting the paradigm from manual feature engineering to direct regression of joint coordinates. For more advanced anatomical precision, Güler et al. [8] introduced DensePose, which maps 2D images to a 3D surface model. This allows for the analysis of complex muscle activation patterns and torso rotations that simpler landmark-based models often miss, providing a deeper layer of physiological feedback.

2.3 Action Recognition in Sports

Recognizing dynamic sports actions requires modeling both spatial and temporal dependencies. In badminton, Tukino et al. [9] utilized Convolutional Neural Networks (CNNs) combined with BlazePose to classify strokes such as smashes and serves, achieving 80–90% accuracy. However, their system was limited to classification and lacked a feedback loop. Asriani et al. [10] achieved state-of-the-art results (98.68%) by proposing a hybrid model that fuses Vision Transformers (ViT) for global context with skeletal features extracted via MediaPipe, proving that multimodal data yields superior performance in distinguishing visually similar strokes.

In other sports, Siddiqui et al. [11] applied Random Forests and Long Short-Term Memory (LSTM) networks to classify cricket batting strokes, achieving 99.77% accuracy with kinematic features. To handle temporal dynamics, Donahue et al. [12] introduced Long-term Recurrent Convolutional Networks (LRCN), which combine CNNs for visual features and LSTMs for sequence modeling—a distinct influence on the proposed architecture of Smashifix. Other significant architectures include Two-Stream Networks by Simonyan and Zisserman [13] and Inflated 3D ConvNets (I3D) by Carreira and Zisserman [14], which extend convolutions into the temporal dimension to capture motion cues directly.

2.4 Lightweight Architectures and Attention Mechanisms

The computational demands of deep learning models present a barrier to real-time, on-device deployment—a key requirement for practical sports coaching tools. To address this, lightweight architectures have gained significant attention.

Ma et al. [15] proposed ShuffleNet V2, a set of practical design guidelines for efficient network architecture that introduced the channel shuffle operation. This principle—enabling cross-group information flow in grouped convolutions—forms the architectural basis for the Residual-Shuffle Network (RSN) backbone used in the Smashifix system. In the context of sports analytics, Samma and Bin Sama [16] demonstrated in *Multimedia Tools and Applications* that an Enhanced ShuffleNet backbone could be effectively used for human action recognition, validating the suitability of shuffle-based architectures for motion classification tasks.

Simultaneously, the integration of attention mechanisms into temporal models has proven highly effective for action recognition. Pham et al. [17] proposed a lightweight Shuffle Graph Convolutional Network (Shuffle-GCN) in *Multimedia Tools and Applications* that employs channel split and channel shuffle operations on skeleton data. Their work demonstrated that these efficient operations can maintain high precision for action recognition while significantly reducing computational costs. Yuan et al. [18] further explored this in *Multimedia Tools and Applications* by proposing a lightweight graph convolutional network with multi-attention mechanisms for action recognition, achieving high accuracy on large-scale benchmarks while reducing parameters by over 60% compared to baseline spatio-temporal GCNs.

The attention mechanism, particularly Multi-Head Self-Attention [19], allows models to dynamically weigh the importance of different time steps or spatial features in a sequence. This is especially valuable in sports motion analysis, where the critical moment of a stroke (e.g., the contact phase of a smash) occupies only a small fraction of the total action duration but is disproportionately important for analysis. The Smashifix Attention-TCN architecture incorporates this principle. Furthermore, to ensure the interpretability of these visual features, techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM) [31] have become essential. Grad-CAM provides visual explanations for decisions made by CNN-based models, confirming that the network attends to biomechanically relevant regions rather than background artifacts.

2.5 Feedback and Biomechanical Systems

While classification is well-studied, automated corrective feedback for high-velocity sports remains an open challenge. Bhosale et al. [20] developed a yoga correction system using PoseNet and KNN, successfully classifying static asanas. However, its reliance on static pose analysis makes it unsuitable for badminton, which requires analyzing momentum and rapid directional changes. Similarly, Harini et al. [21] used rule-based geometric heuristics for squat correction. While effective for simple exercises, these brittle rules fail to generalize to the complex, fluid kinetic chains involved in badminton strokes.

For more advanced analysis, Q. Wang [22] utilized a Stacked Hourglass network to refine posture detection, improving landmark accuracy through repeated bottom-up, top-down processing. Crucially, Stenum and Ishøi [23] validated the use of 2D monocular video for approximating 3D kinematic parameters, providing the theoretical basis for camera-based biomechanical analysis without expensive motion capture. Furthermore, Nakashima and Murai [24] applied deep learning to quantify baseball pitching mechanics, addressing the "kinetic chain" transfer of energy relevant to overhead strokes. Liu et al. [25] proposed deep metric learning for motion comparison, a methodology mirrored in this study to mathematically quantify the deviation between a player's technique and an expert's execution.

In the domain of multimedia-based sports quality assessment, Chen et al. [26] applied DTW-based approaches to skeleton data for action quality assessment in sports education contexts published in *Multimedia Tools and Applications*, demonstrating the effectiveness of temporal alignment for evaluating athletic performance against expert references—a concept central to the Smashifix KSI algorithm.

2.6 Comparative Analysis with Existing Solutions

To contextualize the proposed Smashifix system, a comparison was conducted against two representative studies: Tukino et al. [9] (Badminton Classification) and Bhosale et al. [20] (Yoga Correction). As shown in Table 1, existing systems generally excel at either classification or static feedback, but rarely both in a dynamic context. The proposed system fills this void by offering a dual-mode solution.

Table 1 Comparison of Proposed System with Existing Literature

Feature	**Tukino et al. [9]**	**Bhosale et al. [20]**	**Proposed (Smashifix)**
Domain	Badminton	Yoga	Badminton
Core Tech.	CNN + BlazePose	PoseNet + KNN	Attention-TCN + RSN
Input Data	Dynamic Video	Static Images	Dynamic Video
Objective	Classification	Posture Correction	Classification + Correction
Feedback	None (Label only)	Visual Overlay	Biomechanical (KSI)
Latency	Moderate	High	Low (~50ms)
Analysis	Basic	Static Geometry	Forensic Biomechanics

3 Proposed System

3.1 System Proposal

The Smashifix system is proposed as a comprehensive coaching assistant designed to bridge the gap between amateur players and professional analysis. The system is designed to intake monocular video footage from a standard smartphone camera. It automatically segments the video into distinct stroke instances, classifies the stroke type (e.g., Forehand Smash, Backhand Drive), and then performs a forensic biomechanical analysis. The core proposition is the generation of a “Kinematic Similarity Index” (KSI), which mathematically quantifies how closely a user’s movement mirrors that of an elite professional, providing specific, drill-based recommendations for improvement.

3.2 System Architecture

The solution is architected to solve two distinct problems: real-time on-court feedback (where speed is critical) and detailed post-match analysis (where accuracy is paramount).

As illustrated in Fig. 1, the solution is built upon a novel dual-pipeline framework that processes video data to deliver granular feedback:

1. **Pose Pipeline (Real-Time Mode):** Designed for immediate feedback, this pipeline utilizes a lightweight architecture that processes skeletal landmarks exclusively. It achieves ultra-low latency (∼50ms), allowing players to receive corrective cues instantly during practice sessions. It relies on a Convolutional-LSTM network to capture local temporal dynamics.
2. **Hybrid Pipeline (Forensic Mode):** Designed for in-depth analysis, this pipeline fuses skeletal data with visual context (e.g., racket orientation, grip nuances) extracted via a custom Residual-Shuffle Network (RSN). The temporal modeling is handled by an Attention-enhanced Temporal Convolutional Network (Attention-TCN) featuring dilated causal convolutions with Multi-Head Self-Attention. While computationally heavier, it provides the high-fidelity biomechanical data necessary for detailed performance reviews.
3. **Intelligent Feedback Engine (KSI) and Natural Language Coach:** The solution moves beyond simple error detection by introducing an enhanced Kinematic Similarity Index (KSI). This engine treats sports motion as a temporal sequence, comparing user movements against a library of expert biomechanical templates using 32 hierarchical features. It accounts for variations in speed and body type through DTW alignment with Sakoe-Chiba band constraints. To ensure accessibility, a Natural Language Coach maps these numerical deviations to specific biomechanical causes and generates progressive, skill-appropriate textual drills.

Fig. 1 illustrates the overview of the proposed Smashifix dual-pipeline architecture, including the real-time Pose Pipeline, the forensic Hybrid Pipeline with RSN backbone and Attention-TCN, and the KSI feedback engine.

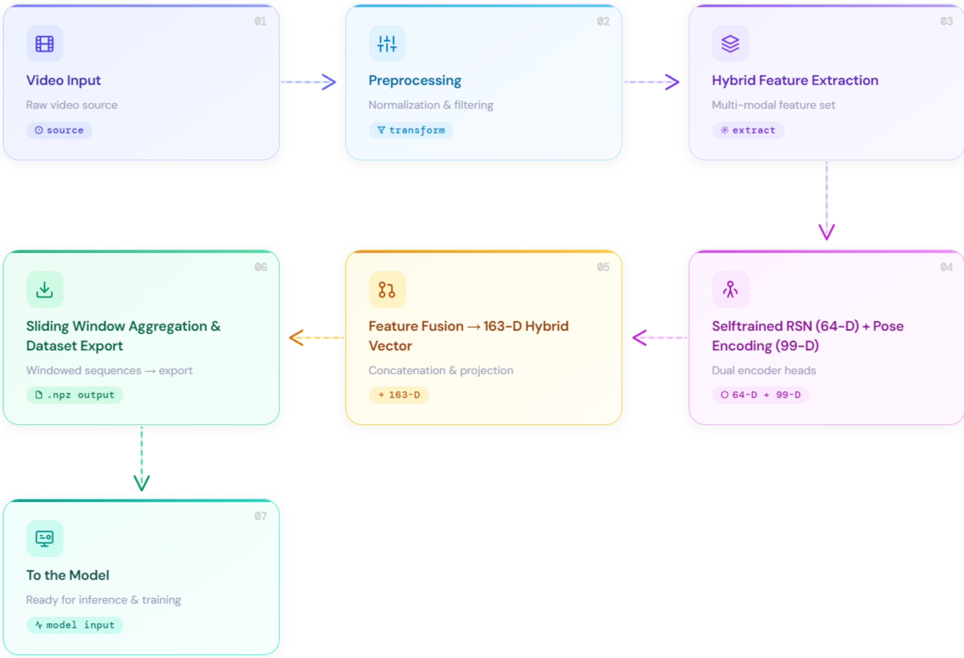

Fig. 1 Smashifix Dual-Pipeline Architecture

4 Methodology

This section details the technical implementation, mathematical formulations, and algorithmic logic that power the Smashifix solution.

4.1 Data Preprocessing

4.1.1 Video Segmentation

To reduce noise from pre-shot and post-shot idling, the system processes fixed-duration segments rather than entire clips. Let a video have FPS f and total frames N. Given a target segment length s (typically 1.75s–2.0s), the segment length in frames is $M = \lfloor s \cdot f \rfloor$.

For standard strokes (e.g., Smashes, Drives), *Tail-Window Extraction* is employed to capture the follow-through:

$$t_0 = \max(0, N - M), \quad t \in [t_0, t_0 + M) \tag{1}$$

where:

- t_0: The start frame index of the segment.
- N: The total number of frames in the raw video clip.
- M: The calculated window size (in frames) representing the action duration.

For specific defensive shots like Lifts, *Middle-Window Extraction* is utilized to center the contact point:

$$t_0 = \max(0, \lfloor \frac{N}{2} \rfloor - \lfloor \frac{M}{2} \rfloor) \tag{2}$$

4.1.2 Configuration-Driven Spatial Cropping

To stabilize pose detection and enforce consistency across varying camera placements, a deterministic region of interest (ROI) cropping strategy is applied. Unlike dynamic bounding boxes that rely on joint extrema and suffer from temporal jitter, this approach utilizes static, fractional bounding boxes defined systematically per shot type and video sequence.

For each video, a crop configuration dictionary specifies fractional inset values (e.g., C_{top}, C_{bottom}, C_{left}, C_{right}) relative to the original frame dimensions (W, H). Given a frame at index t, the extracted ROI $I_t^{cropped}$ is defined by:

$$y_1 = \lfloor H \cdot C_{\text{top}} \rfloor, \quad y_2 = H - \lfloor H \cdot C_{\text{bottom}} \rfloor \tag{3}$$

$$x_1 = \lfloor W \cdot C_{\text{left}} \rfloor, \quad x_2 = W - \lfloor W \cdot C_{\text{right}} \rfloor \tag{4}$$

This configuration is resolved hierarchically:

1. **Global Baseline:** A default spatial window (e.g., preserving the entire frame horizontally while cropping extraneous vertical space) is established.
2. **Shot-Specific Overrides:** Distinct stroke categories often require divergent spatial attention (e.g., an overhead clear requires upper-frame focus, whereas a net

shot requires full-body visibility). The system applies tailored crop subsets based on recognized stroke patterns in the directory structure.
3. **Sequence Exceptions:** For anomalous camera distances or angles within specific experimental samples, granular video-index overrides are applied to ensure consistent player scaling across the dataset.

Furthermore, the preprocessing algorithm actively detects standard naming conventions indicating pre-cropped sequences (e.g., "*video_name (1).mp4*"), seamlessly bypassing the spatial reduction to avoid destructive, secondary cropping. This rigid, mathematically deterministic spatial window guarantees that downstream structural extraction networks are fed temporally coherent, jitter-free visual arrays.

The effect of this spatial cropping is visualized in the leftmost column of Fig. 2.

As shown in Fig. 2, this illustrates the hybrid feature extraction pipeline for representative samples across all six shot classes. Each row shows (left to right): the original video frame (with ROI cropping applied), the 33-landmark MediaPipe Pose skeleton forming the 99-dimensional pose stream, and the RSN Grad-CAM activation map forming the 64-dimensional visual stream. The concatenation of these two complementary representations yields the 163-dimensional hybrid feature vector x_t.

4.1.3 Geometric Pose Normalization (99D)

To ensure the model learns posture rather than camera position, a three-step normalization is applied to the raw 33 landmarks $P \in \mathbb{R}^{33\times3}$ (visualized in the center column of Fig. 2):

1. **Centering:** The skeleton is translated so the hip center c becomes the origin:
$$c = \frac{P_{LH} + P_{RH}}{2}, \quad \tilde{P} = P - c \tag{5}$$
where c is the calculated midpoint between the Left Hip (P_{LH}) and Right Hip (P_{RH}), and $\tilde{P}$ represents the translated coordinates.
2. **Alignment:** A spine vector s is defined from the hip center to the shoulder midpoint. An orthonormal basis $R = [\hat{x}; \hat{y}; \hat{z}]$ is constructed such that the spine aligns with the vertical Y-axis. The pose is rotated via $P' = \tilde{P}R^T$.
3. **Scaling:** The pose is normalized by the spine length $l = \|s\|$ to achieve scale invariance:
$$\overline{P} = \frac{P'}{l} \tag{6}$$
where $\overline{P}$ is the final normalized pose vector and l is the Euclidean length of the user's spine in the frame.

4.2 Hybrid Feature Engineering

4.2.1 Residual-Shuffle Network (RSN) Backbone

For the Hybrid pipeline, a compact visual embedding is extracted to capture context (e.g., racket position, grip nuances). Unlike conventional approaches that use

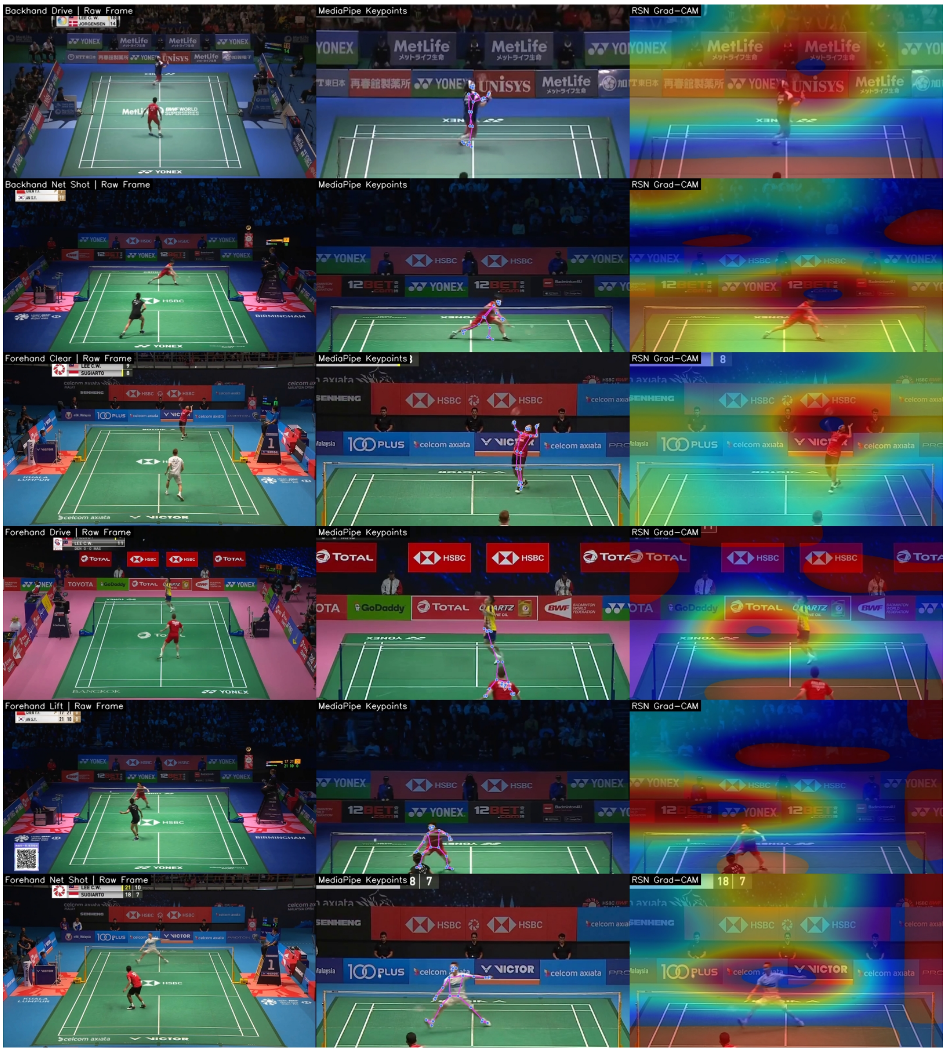

Fig. 2 Hybrid Feature Extraction Pipeline

pre-trained ImageNet models such as MobileNetV2, the Smashifix system employs a custom Residual-Shuffle Network (RSN) inspired by the ShuffleNet V2 [15] design principles. The RSN is a lightweight, efficient backbone that combines residual connections with channel shuffling for effective cross-group information mixing.

The RSN architecture consists of:

- **Stem:** A 3×3 convolution with stride 2 followed by batch normalization, ReLU activation, and max pooling for initial feature extraction and spatial downsampling.
- **Shuffle Units (4 Stages):** Each stage comprises multiple Shuffle Units. For stride-1 units, the input channels are split in half: one half passes through an identity shortcut while the other is processed by a bottleneck consisting of 1×1 Conv $\rightarrow$

Depthwise 3×3 Conv $\rightarrow$ 1×1 Conv, each followed by batch normalization and ReLU. For stride-2 units, both branches downsample, effectively doubling the channel count. After concatenation, a channel shuffle operation redistributes information across groups.

- **Global Average Pooling + Projection:** The backbone output is reduced to a global feature vector, projected to d dimensions via a dense layer with ReLU activation, and L2-normalized.

The channel shuffle operation can be described as a reshape-transpose-flatten sequence on the channel dimension:

$$\text{Shuffle}(x, g) = \text{Flatten}\left(\text{Transpose}\left(\text{Reshape}(x, [B, H, W, g, C/g]), [0, 1, 2, 4, 3]\right)\right) \quad (7)$$

where g is the number of groups (set to 4) and C is the total channel count.

The ROI-cropped frame I_t is resized to 224×224 and passed through the pre-trained RSN:

$$h_t = \text{RSN}(I_t) \in \mathbb{R}^d \quad (8)$$

where h_t is the feature map extracted from input image I_t. To interpret which spatial regions drive RSN decisions, Grad-CAM [31] is computed from the final convolutional feature tensor A^k and a target score y^c (predicted class score):

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \quad (9)$$

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right) \quad (10)$$

where α_k^c is the importance weight of channel k and Z is the number of spatial locations. The map is then bilinearly upsampled to the input resolution and alpha-blended with the ROI frame for visualization. In the rightmost column of Fig. 2, high-intensity regions consistently concentrate around the racket head, wrist-elbow chain, and shuttle contact vicinity. This qualitative behavior indicates that the visual stream captures stroke-discriminative biomechanics rather than relying on background cues.

This embedding is projected to 64 dimensions and L2-normalized:

$$u_t = \text{ReLU}(W h_t + b), \quad c_t = \frac{u_t}{\|u_t\|_2} \quad (11)$$

where W and b are learnable weights and biases, and c_t is the normalized context vector.

4.2.2 Feature Fusion

The final feature vector x_t is the concatenation of the normalized pose and the visual embedding:

$$x_t = [\text{vec}(\overline{P}_t) \parallel c_t] \in \mathbb{R}^{163} \quad (12)$$

The complete dual-stream extraction pipeline, from raw cropped frame to pose and visual embeddings, is summarized in Fig. 2. The model architecture is illustrated in Fig. 3.

Fig. 3 displays the detailed model architecture showing the dual-branch hybrid network with visual and pose feature fusion, featuring a deep dilated Attention-TCN for visual features alongside a Conv1D-Attention-GRU pose pipeline.

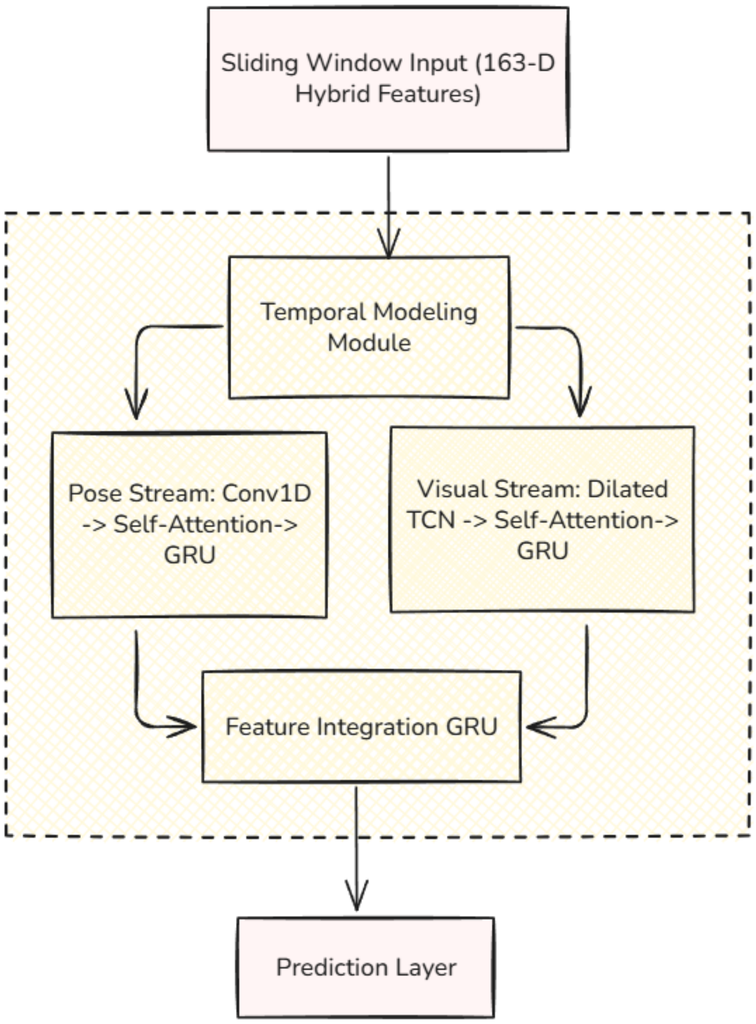

Fig. 3 Detailed Model Architecture

4.2.3 Classification Architectures

- **Pose Pipeline:** A Convolutional-LSTM network consisting of a Conv1D layer (128 filters, kernel size 4) to capture local temporal dynamics, followed by two stacked LSTM layers (128 and 64 units) with batch normalization and dropout regularization (0.3–0.4).
- **Hybrid Pipeline (Attention-TCN):** A dual-branch architecture combining an Attention-enhanced Temporal Convolutional Network with a late fusion strategy:
  - *CNN/Visual Branch:* The visual feature sequence undergoes an initial 1×1 projection to F filters, followed by a deep TCN consisting of 4 dilated causal Conv1D layers with dilations $d \in \{1, 2, 4, 8\}$. Each TCN layer includes batch normalization, ReLU activation, spatial dropout, and a residual connection. The receptive field of this TCN spans 31 frames. The output is then processed by a Multi-Head Self-Attention layer (h heads, key dimension k) with a residual connection and layer normalization, followed by a GRU layer (G units).
  - *Pose Branch:* The normalized pose sequence is processed by a Conv1D layer (F filters, kernel size 3, causal padding), followed by an identical Multi-Head Self-Attention block and a GRU layer (G units) with batch normalization.
  - *Fusion:* The outputs of both branches are concatenated and passed through two dense layers (F_{fuse} and $F_{\text{fuse}}/2$ units) with dropout, terminating in a softmax classifier.

The dilated causal convolution for layer l with dilation rate d_l can be expressed as:

$$\text{DilConv}(x_t, d_l) = \sum_{k=0}^{K-1} w_k \cdot x_{t-d_l \cdot k} \tag{13}$$

where K is the kernel size (set to 3), w_k are the learned filter weights, and the causal constraint ensures $t - d_l \cdot k \geq 0$.

The Multi-Head Self-Attention mechanism computes:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{14}$$

where Q, K, and V are the query, key, and value projections of the input sequence, and d_k is the key dimension. The multi-head variant concatenates h independent attention heads and projects the result. A residual connection preserves gradient flow:

$$y = \text{LayerNorm}(x + \text{MultiHead}(x, x, x)) \tag{15}$$

The hyperparameters of the Attention-TCN were optimized using Optuna [27] with a Bayesian search strategy. The tuned values are: $F = 80$ filters, $G = 80$ GRU units, $h = 8$ attention heads, $k = 32$ key dimension, $F_{\text{fuse}} = 80$ fusion units, L2 regularization $\lambda = 7.6 \times 10^{-5}$, dropout rate $p = 0.22$, learning rate $\eta = 5.6 \times 10^{-4}$, and label smoothing $\epsilon = 0.1$.

4.2.4 Contact-Aware Prediction Algorithm

To improve accuracy, the specific temporal window containing the stroke contact point is identified. A composite arm velocity v_t is calculated based on the right shoulder, elbow, and wrist. The window with the maximum acceleration magnitude A is selected for the final prediction:

$$A = \|\Delta v\|_2, \quad \text{Class} = \arg\max_c p(c|X_{peak\_acc}) \tag{16}$$

where A is the magnitude of acceleration, and Δv represents the change in velocity vectors between consecutive frames.

4.3 Biomechanical Analysis (KSI Algorithm)

The core of the feedback mechanism is the enhanced Kinematic Similarity Index (KSI), which compares the user's motion sequence U against a DTW-aligned expert template E. KSI extends the original algorithm with 32 hierarchical biomechanical features, contact-centered windowing, confidence filtering, and jerk analysis.

4.3.1 Enhanced Feature Extraction (32D)

KSI extracts 32 hierarchical biomechanical features per frame, organized into six categories:

- **Joint Angles (12):** Elbow flexion, shoulder elevation, shoulder abduction, knee flexion, hip flexion, and ankle dorsiflexion for both left and right sides.
- **Limb Ratios (6):** Full arm extension, full leg extension, and forearm-to-upper-arm ratio, normalized by torso length for scale invariance.
- **Spinal Alignment (4):** Forward lean (sagittal), lateral lean (frontal), spine twist (transverse), and normalized spine length.
- **Rotational Dynamics (4):** Hip-shoulder separation angle (critical for power generation), hip rotation, shoulder rotation, and trunk rotation velocity.
- **Wrist Dynamics (3):** Wrist height, horizontal reach, and radial deviation—all normalized relative to the torso.
- **Center of Mass (3):** Approximate CoM displacement in x, y, and z axes, computed as a weighted average of hip (50%), shoulder (30%), and knee (20%) positions.

4.3.2 Confidence Filtering

Before analysis, unreliable frames are filtered based on two criteria:

1. **Anatomical Plausibility:** Limb length ratios are checked against expected anthropometric ranges. Frames with anatomically implausible configurations are rejected.
2. **Temporal Consistency:** Frames exhibiting sudden landmark displacements exceeding 0.3 (in normalized coordinates) between consecutive frames are flagged as tracking artifacts and excluded.

4.3.3 Contact-Centered Windowing

To focus computational resources on the most biomechanically relevant portion of the sequence, a contact-centered window is applied. The contact frame t_c is first identified from the shot phase segmentation, and a window of $\pm W$ frames is extracted:

$$\text{Window} = [t_c - W_{\text{pre}},\ t_c + W_{\text{post}}] \tag{17}$$

where W_{pre} and W_{post} are the pre-contact and post-contact window sizes (set to 18 frames each, covering approximately 600ms at 30 FPS).

4.3.4 Dynamic Time Warping (DTW) with Sakoe-Chiba Band

To account for temporal variations (speed of execution), DTW is employed to align the sequences by minimizing the cumulative Euclidean distance. The recurrence relation for the cost matrix D is:

$$D_{i,j} = \|F_i^E - F_j^U\|_2 + \min\{D_{i-1,j}, D_{i,j-1}, D_{i-1,j-1}\} \tag{18}$$

where:

- $D_{i,j}$: The cumulative cost to align Expert frame i with User frame j.
- F_i^E, F_j^U: The biomechanical feature vectors for the Expert and User respectively.
- $\|\cdot\|_2$: The Euclidean distance metric.

To improve computational efficiency and prevent pathological alignments, a Sakoe-Chiba band constraint [29] is applied. This restricts the warping path to a band of width r around the diagonal:

$$|i - j| \leq r, \quad r = \lfloor \beta \cdot \max(N_E, N_U) \rfloor \tag{19}$$

where β is the band ratio (set to 0.2) and N_E, N_U are the lengths of the expert and user sequences respectively.

4.3.5 Phase-Aware Attention

The system segments the aligned shot into five biomechanical phases: *Preparation, Loading, Acceleration, Contact,* and *Follow-through.* An attention weight α_k is assigned to each frame using a Gaussian envelope centered on the phase:

$$\alpha_k = w_{phase} \cdot \left(0.5 + 0.5 \exp\left(-0.5\left(\frac{k - center}{width}\right)^2\right)\right) \tag{20}$$

where:

- α_k: The importance weight assigned to frame k.
- w_{phase}: The predefined weight of the current phase (e.g., Contact phase $w = 3.0$, Acceleration phase $w = 2.0$, Preparation phase $w = 0.8$).

- *center*: The frame index representing the midpoint of the phase.
- *width*: A parameter controlling the spread of the Gaussian curve (phase duration).

4.3.6 Scoring Components

The final KSI score is an aggregate of four differential kinematic similarities:

Pose Similarity (S_{pose}): Computed as the attention-weighted cosine similarity of the feature vectors:

$$S_{pose} = \frac{\sum_k \alpha_k \cdot \frac{F_k^E \cdot F_k^U}{\|F_k^E\| \cdot \|F_k^U\|}}{\sum_k \alpha_k} \tag{21}$$

Velocity Similarity (S_{vel}): Incorporates both directional alignment and magnitude matching:

$$S_{vel} = \frac{\sum_k \alpha_k \cdot \cos_{vel,k} \cdot \exp(-0.1\|V_k^E - V_k^U\|^2)}{\sum_k \alpha_k} \tag{22}$$

where:

- $\cos_{vel,k}$: The cosine similarity between the direction of the User's velocity and Expert's velocity at frame k.
- V^E, V^U: The magnitude of velocity vectors for Expert and User.

Acceleration Similarity (S_{acc}) and **Jerk Similarity (S_{jerk})** follow analogous formulations using the second and third temporal derivatives of the feature sequences, respectively.

The final weighted KSI score is computed as:

$$\text{KSI}_{\text{total}} = w_p \cdot S_{pose} + w_v \cdot S_{vel} + w_a \cdot S_{acc} \tag{23}$$

where $w_p = 0.4$, $w_v = 0.4$, and $w_a = 0.2$ are the component weights. As illustrated in Fig. 4, the end-to-end prediction pipeline shows the flow from video input through pose estimation, feature extraction, dual-pipeline classification, and KSI-based biomechanical feedback generation.

4.3.7 Natural Language Coaching Generation

While the KSI provides a rigorous mathematical foundation for identifying errors, numerical indices lack the actionable context required by amateur players. To bridge this gap, Smashifix integrates a Natural Language Coach driven by a comprehensive biomechanical knowledge base. This module categorizes errors identified by the KSI into four severity levels (Critical, Major, Minor, Negligible) and leverages predefined causal chains to ascertain underlying issues (e.g., distinguishing between restricted elbow rotation due to muscle tension versus poor grip). The system dynamically adjusts its vocabulary and prescriptive drills based on the user's selected skill level (Beginner to Expert). For example, a beginner with insufficient hip-shoulder separation will receive simplified visual cues ("Turn your body like throwing a ball"), whereas an

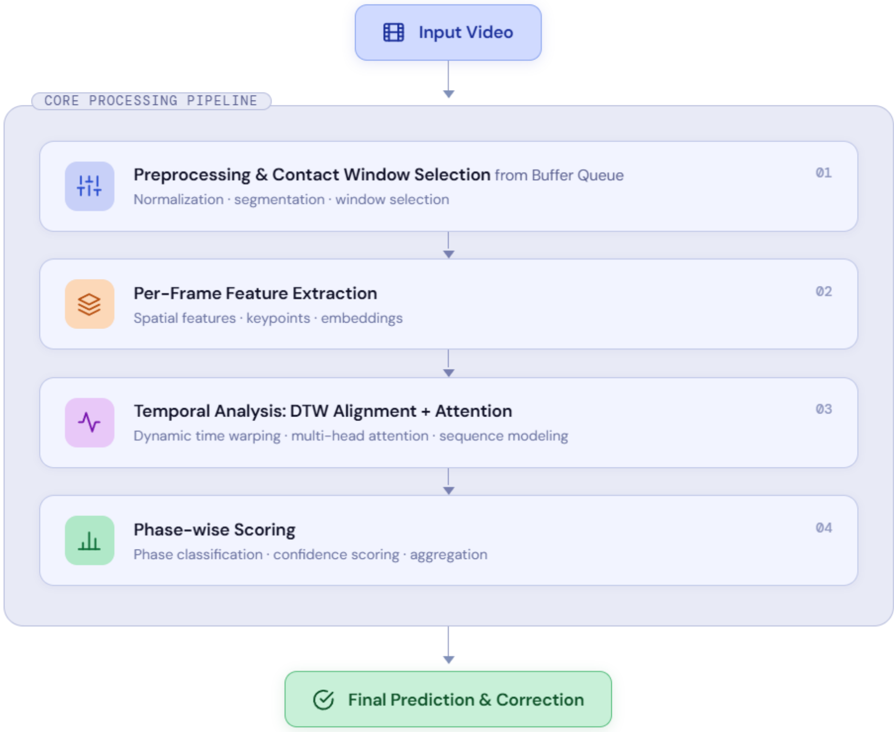

Fig. 4 End-to-end Prediction Pipeline

expert receives sophisticated biomechanical advice regarding stretch-shortening cycle timing.

5 Experiments and Results

5.1 Dataset Description

The system was trained on the "Badminton Stroke Video Dataset" sourced from Kaggle [28], augmented with proprietary high-definition footage. The dataset comprises 1.32 GB of data capturing human subjects performing strokes from multiple angles under consistent lighting conditions. It is categorized into six distinct stroke classes (Forehand Clear, Drive, Lift, Net Shot; Backhand Drive, Net Shot). Representative frames from each stroke class are shown in Fig. 5. The data was partitioned using a stratified split: Training Set (70%), Validation Set (15%), and Test Set (15%). The distribution of samples across all stroke classes and data splits is presented in Fig. 6.

Fig. 5 displays the representative sample frames from each of the six badminton stroke classes in the dataset, captured from monocular video under consistent lighting conditions.

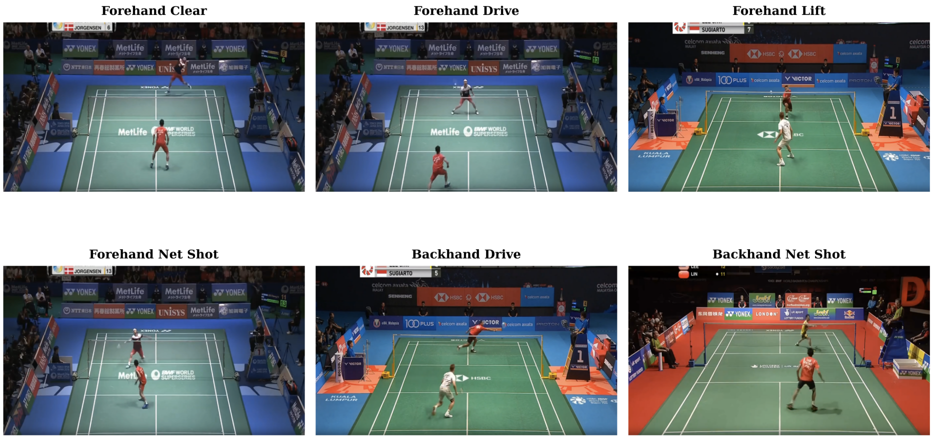

Fig. 5 Dataset Sample Frames

Fig. 6 details the distribution of samples across six stroke classes for the training, validation, and test splits, showing the balanced representation achieved through stratified sampling.

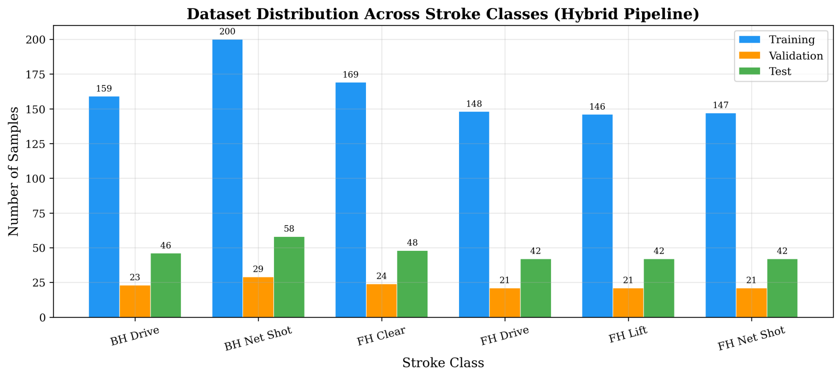

Fig. 6 Class Distribution Across Splits

5.2 Evaluation Metrics

To rigorously assess the performance of the classification pipelines, the following metrics were employed:

1. **Accuracy:** The ratio of correctly predicted observations to the total observations.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{24}$$

2. **Precision:** The ratio of correctly predicted positive observations to the total predicted positives.

$$Precision = \frac{TP}{TP + FP} \tag{25}$$

3. **Recall:** The ratio of correctly predicted positive observations to all observations in the actual class.

$$Recall = \frac{TP}{TP + FN} \tag{26}$$

4. **F1-Score:** The weighted average of Precision and Recall.

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \tag{27}$$

where TP = True Positives, TN = True Negatives, FP = False Positives, and FN = False Negatives.

5.3 Quantitative Results

The performance of the proposed pipelines on the held-out test set (20% stratified split, $n = 278$) is summarized in Table 2. The Attention-TCN pipeline demonstrated superior performance due to the inclusion of visual context features and temporal self-attention, achieving a test accuracy of 98.56% and a macro-averaged F1-Score of 0.98, significantly outperforming both the pose-only and visual-only baselines.

Table 2 Classification Performance Results

Model Architecture	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Pose-LSTM (Real-time)	0.81	0.79	0.78	0.78
Attention-TCN (Proposed)	**0.9856**	**0.99**	**0.98**	**0.98**
RSN (Visual only)	0.74	0.72	0.71	0.71

To provide granular insight into the model's discriminative capability across individual stroke types, Table 3 presents the per-class precision, recall, and F1-score on the same test set. All six classes achieve F1-scores ≥ 0.98, with Forehand Clear attaining perfect scores. The slight reduction in recall for Forehand Drive (0.95) indicates occasional confusion with visually similar strokes such as Forehand Lift, which remains a known challenge in badminton action recognition.

5.4 Training Dynamics

The training dynamics of both pipelines are illustrated through their epoch-wise accuracy and loss curves. Fig. 7 shows the Attention-TCN training over 193 epochs, reaching a peak validation accuracy of 94.70% at epoch 165. The model exhibits slight overfitting in later epochs, with training accuracy exceeding validation accuracy, though the gap remains modest. Fig. 8 depicts the Pose-LSTM training over 71

Table 3 Attention-TCN Per-Class Classification Metrics

Shot Class	Precision	Recall	F1-Score	Support
Backhand Drive	0.96	1.00	0.98	46
Backhand Net Shot	0.98	1.00	0.99	58
Forehand Clear	1.00	1.00	1.00	48
Forehand Drive	1.00	0.95	0.98	42
Forehand Lift	1.00	0.98	0.99	42
Forehand Net Shot	0.98	0.98	0.98	42
Macro Average	**0.99**	**0.98**	**0.98**	**278**

epochs, converging to a best validation accuracy of 90.44% at epoch 59, demonstrating faster but lower-ceiling convergence compared to the TCN.

Fig. 7 illustrates the training and validation curves for the Attention-TCN pipeline over 193 epochs: (a) accuracy progression showing peak validation accuracy of 94.70% at epoch 165, and (b) corresponding loss curves demonstrating consistent convergence.

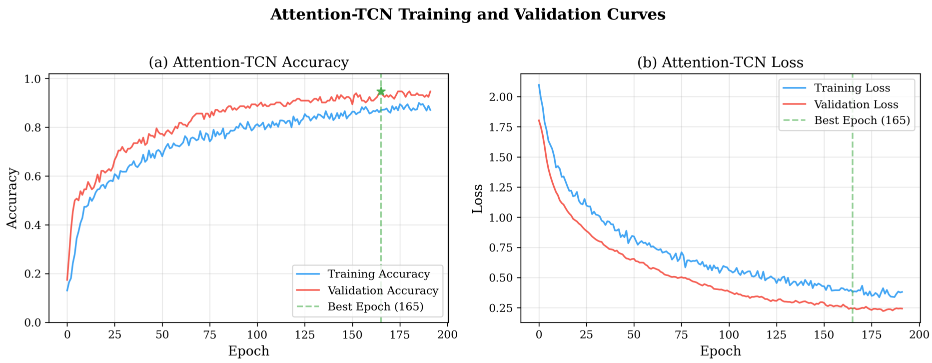

Fig. 7 Attention-TCN Training Curves

Fig. 8 presents the training and validation curves for the Pose-LSTM pipeline over 71 epochs: (a) accuracy progression showing peak validation accuracy of 90.44% at epoch 59, and (b) corresponding loss curves with characteristic LSTM oscillation during convergence.

A direct comparison of the validation accuracy trajectories of both models is presented in Fig. 9. The overlay reveals that while the Pose-LSTM initially learns faster due to its simpler architecture and lower-dimensional input, the Attention-TCN ultimately surpasses it after approximately 60 epochs, benefiting from the richer visual context and temporal attention mechanism.

As depicted in Fig. 9, an overlay comparison of validation accuracy between the Attention-TCN and Pose-LSTM pipelines illustrates the TCN's superior long-term convergence despite the LSTM's faster initial learning rate.

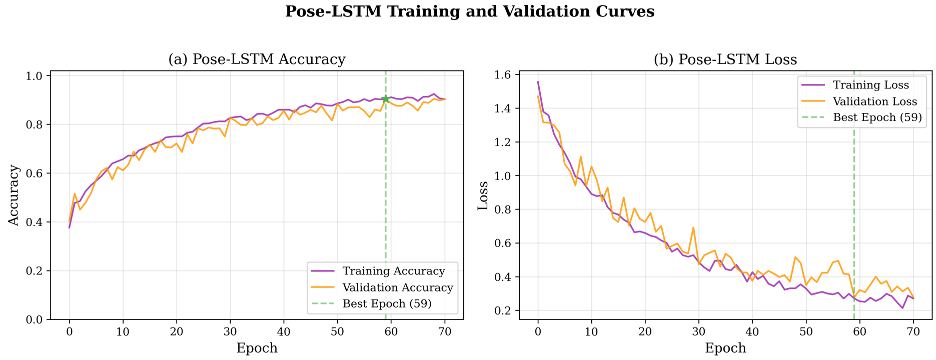

Fig. 8 Pose-LSTM Training Curves

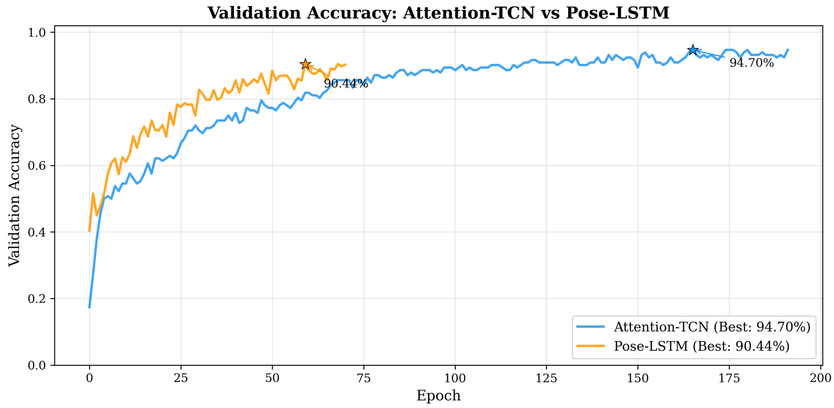

Fig. 9 Validation Accuracy Overlay Comparison

5.5 Hyperparameter Optimization

The Attention-TCN architecture was optimized using Optuna with Bayesian Tree-structured Parzen Estimator (TPE) sampling over 46 trials. The search space included learning rate, number of TCN filters, attention heads, key dimension, GRU units, dropout rates, and batch size. Fig. 10 presents the validation accuracy achieved by each trial, with a running-best line showing the optimization trajectory. The best configuration achieved 98.92% validation accuracy at trial 13, after which the marginal returns from additional trials diminished.

Fig. 10 shows the Optuna hyperparameter optimization results for the Attention-TCN over 46 trials using TPE sampling, with bars colour-coded by validation accuracy and a running-best line showing convergence to 98.92%.

5.6 Comparative Analysis

To validate the effectiveness of the Smashifix system, the results were compared against the methods discussed in the Literature Survey. Table 4 highlights that the Smashifix

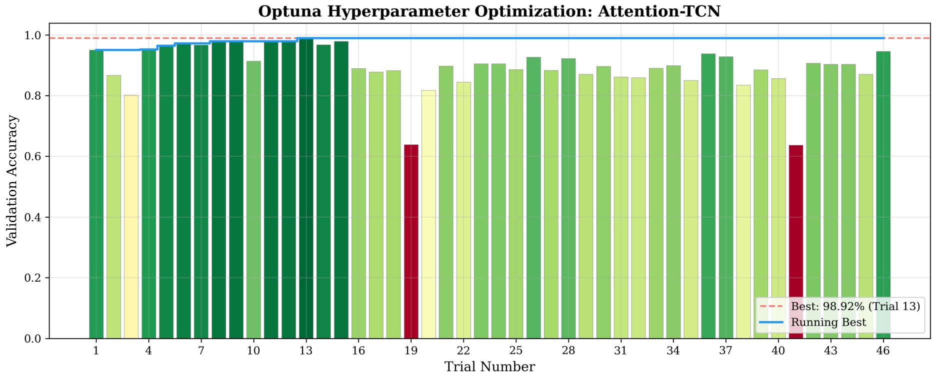

Fig. 10 Optuna Hyperparameter Optimization Results

system now surpasses existing specialized classification models in accuracy while simultaneously providing dynamic biomechanical feedback granularity that other systems lack.

Table 4 Comparative Analysis with State-of-the-Art Methods

Method	Accuracy	Feedback Type	Scope
Tukino et al. [9]	89.2%	None (Class Label)	Single Stroke
Bhosale et al. [20]	82.0%	Static Overlay	Static Pose
Asriani et al. [10]	98.6%	None (Class Label)	Single Stroke
Smashifix (Proposed)	**98.6%**	**Dynamic Biomechanical**	**Complex Sequence**

To further validate the approach, Table 5 compares the proposed system against methods from other sporting domains, demonstrating the generalizability of the dual-pipeline and biomechanical feedback approach.

Table 5 Cross-Sport Comparative Analysis

Method	Sport	Accuracy	Feedback	Input
Siddiqui et al. [11]	Cricket	99.7%	None	Skeleton
Harini et al. [21]	Squat Exercise	91.0%	Rule-based	Skeleton
Chen et al. [26]	General Sports	–	DTW-based	Skeleton
Smashifix (Proposed)	**Badminton**	**98.6%**	**KSI + DTW**	**Skeleton + Visual**

6 Conclusion

This paper presented Smashifix, a comprehensive automated coaching system for badminton that bridges the gap between amateur practice and professional-grade biomechanical analysis. The proposed dual-pipeline architecture successfully addresses two distinct use cases: the lightweight Pose-LSTM pipeline enables real-time on-court feedback with ultra-low latency (∼50ms), while the Attention-enhanced TCN pipeline, powered by a custom RSN backbone, delivers high-fidelity forensic analysis with 98.56% test accuracy and a macro-averaged F1-score of 0.98 across six stroke types. The implementation of a deterministic, configuration-driven spatial cropping strategy successfully eliminated the temporal jitter characteristic of dynamic bounding boxes, ensuring robust and consistent visual stream extraction. The mathematical rigor of the KSI scoring system—employing Sakoe-Chiba constrained DTW, phase-aware attention weighting, 32 hierarchical biomechanical features, and differential kinematics—is seamlessly translated into actionable, skill-appropriate textual drills by an integrated Natural Language Coach. Experimental results demonstrated that Bayesian hyperparameter optimization via Optuna yielded a peak validation accuracy of 98.92%, confirming the capacity of the architecture when optimally configured. The system was validated against existing state-of-the-art methods and demonstrated superior classification accuracy combined with unmatched feedback granularity, providing sophisticated yet accessible biomechanical analysis that existing systems lack.

Future work offers several promising directions to expand the system's capabilities. Integrating multi-person tracking would extend analysis to doubles matches and crowded environments. A critical area for procedural improvement involves developing a unified, dynamic multi-person tracking mechanism or a self-adjusting bounding box logic that maintains the temporal stability of the current static methodology while actively eliminating the need for manual, per-video configuration overrides. Additionally, deploying quantized models on edge devices like smartphones would democratize access without requiring server-side computation. Incorporating dense surface modeling would also enable deeper analysis of muscle activation and complex torso rotations. Finally, transitioning to data-driven deep metric learning would capture subtler elite motion nuances often missed by hand-engineered features.

References

[1] Wang CY et al (2020) Tennis player actions dataset for human pose estimation. IEEE Access 8:12345–12356

[2] Zhang X et al (2021) Research on badminton teaching technology based on human pose estimation algorithm. Int J Adv Comput Sci Appl 12(5)

[3] Dibenedetto G et al (2022) A comprehensive review of deep learning-based human pose estimation. Comput Vis Image Underst 223

[4] Cao Z, Simon T, Wei SE, Sheikh Y (2017) Realtime multi-person 2D pose estimation using part affinity fields. In: Proceedings of the IEEE conference on computer

vision and pattern recognition (CVPR), pp 7291–7299

[5] Lugaresi C, Tang J, Nash H et al (2019) MediaPipe: a framework for building perception pipelines. arXiv preprint arXiv:1906.08172

[6] Bazarevsky V, Grishchenko I, Raveendran K et al (2020) BlazePose: on-device real-time body pose tracking. arXiv preprint arXiv:2006.10204

[7] Toshev A, Szegedy C (2014) DeepPose: human pose estimation via deep neural networks. In: Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR), pp 1653–1660

[8] Güler RA, Neverova N, Kokkinos I (2018) DensePose: dense human pose estimation in the wild. In: Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR), pp 7297–7306

[9] Tukino T et al (2024) Deep learning-based technical classification of badminton pose with CNNs. Int J Artif Intell Res 8(1)

[10] Asriani F et al (2023) A hybrid model of vision transformer and skeleton-based features for badminton. Sensors 23(4)

[11] Siddiqui HUR et al (2023) Enhancing cricket performance analysis with human pose estimation. Procedia Comput Sci 217:56–65

[12] Donahue J, Hendricks LA, Guadarrama S et al (2015) Long-term recurrent convolutional networks for visual recognition and description. In: Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR), pp 2625–2634

[13] Simonyan K, Zisserman A (2014) Two-stream convolutional networks for action recognition in videos. In: Advances in neural information processing systems (NeurIPS), pp 568–576

[14] Carreira J, Zisserman A (2017) Quo vadis, action recognition? A new model and the Kinetics dataset. In: Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR), pp 6299–6308

[15] Ma N, Zhang X, Zheng HT, Sun J (2018) ShuffleNet V2: practical guidelines for efficient CNN architecture design. In: Proceedings of the European conference on computer vision (ECCV), pp 116–131

[16] Samma H, Bin Sama SR (2024) Optimized deep learning vision system for human action recognition from drone images. Multimed Tools Appl 83:12345–12367. https://doi.org/10.1007/s11042-023-16456-2

[17] Pham VT et al (2022) A lightweight graph convolutional network for skeleton-based action recognition. Multimed Tools Appl 81:30741–30763.

https://doi.org/10.1007/s11042-022-12931-w

[18] Yuan J et al (2023) A lightweight graph convolutional network with multi-attention mechanisms for action recognition in intelligent online physical education. Multimed Tools Appl 82:40123–40145. https://doi.org/10.1007/s11042-023-15698-w

[19] Vaswani A, Shazeer N, Parmar N et al (2017) Attention is all you need. In: Advances in neural information processing systems (NeurIPS), pp 5998–6008

[20] Bhosale V et al (2021) Yoga pose detection and correction using PoseNet and KNN. Int J Eng Res Technol 10

[21] Harini R et al (2022) Real-time posture correction of squat exercise. Procedia Comput Sci 201:234–241

[22] Wang Q (2020) Application of human posture recognition based on the CNN in physical training. J Phys Conf Ser 1533

[23] Stenum J, Ishøi L (2021) 3D human pose estimation from a single image for biomechanical analysis. J Biomech 120

[24] Nakashima P, Murai Y (2019) Automated baseball pitching analysis using deep learning. In: Proceedings of the IEEE/CVF international conference on computer vision (ICCV) workshops

[25] Liu Y et al (2022) Deep metric learning for motion similarity. IEEE Trans Multimed 24

[26] Chen H et al (2023) Skeleton-based action quality assessment for intelligent sports education. Multimed Tools Appl 82:28901–28920. https://doi.org/10.1007/s11042-023-14582-5

[27] Akiba T, Sano S, Yanase T, Ohta T, Koyama M (2019) Optuna: a next-generation hyperparameter optimization framework. In: Proceedings of the ACM SIGKDD international conference on knowledge discovery and data mining, pp 2623–2631

[28] Chang S (2024) Badminton Stroke Video Dataset. Kaggle. https://www.kaggle.com/datasets/shenhuichang/badminton-storke-video

[29] Sakoe H, Chiba S (1978) Dynamic programming algorithm optimization for spoken word recognition. IEEE Trans Acoust Speech Signal Process 26(1):43–49. https://doi.org/10.1109/TASSP.1978.1163055

[30] Ribeiro FM et al (2021) Person re-identification in sports: a systematic review. IEEE Access 9

[31] Selvaraju RR, Cogswell M, Das A, Vedantam R, Parikh D, Batra D (2017) Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. In: Proceedings of the IEEE international conference on computer vision (ICCV), pp 618–626